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# Creating Electronic Books with Code and Artificial Intelligence Assistants

Course for the Faculties of Sciences and Chemical Sciences

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Welcome to **Creating Electronic Books with Code and Artificial Intelligence Assistants**.

## What is this?

This is the course material and a template designed for the teaching staff of the **Faculties of Sciences and Chemical Sciences at USAL** to create interactive teaching books easily. The main idea of this template is for it to be broad enough to cover many use cases and for each student to adapt it to their own course, while also being sufficiently equipped to be used easily with the help of AI assistants (such as GitHub Copilot, Gemini, Claude, Codex, etc.) and a code editor such as VS Code.

## Content

In this book you will find:

- *Tutorials* to learn how to use the template
- *Examples by Degree* to see real cases
- Information on *how to cite* and *licenses*

## PDF version

You can also download the printable version of the book:

- Download PDF in English
- Descargar PDF en español

### Note

This project is designed to be used with **VS Code** and **AI** assistants.



**Part I**

**Sample**





## CENTRAL LIMIT THEOREM

The **Central Limit Theorem** (CLT) is a fundamental result in probability theory and statistics. It states that, given certain conditions, the sample mean of a sufficiently large number of independent and identically distributed (i.i.d.) random variables will be approximately normally distributed, regardless of the underlying distribution of the original population.

### 1.1 Mathematical Expression

Let  $X_1, X_2, \dots, X_n$  be a sequence of i.i.d. random variables drawn from a population with expected value  $\mu$  and finite variance  $\sigma^2$ . As the sample size  $n$  increases, the distribution of the sample mean  $\bar{X}_n$  approaches a normal distribution.

The standardization of the sum of these variables is expressed by the following equation:

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \quad (1.1)$$

As  $n \rightarrow \infty$ , the random variable  $Z_n$  converges in distribution to the standard normal distribution  $\mathcal{N}(0, 1)$ :

$$\lim_{n \rightarrow \infty} P(Z_n \leq z) = \Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt \quad (1.2)$$

#### Important Note

The CLT is the reason why the normal distribution is so prevalent in nature and in statistical analysis. It allows making statistical inferences about population means even if the original distribution is not normal, provided the sample is sufficiently large (typically  $n \geq 30$ ).



# **Part II**

# **Information**



## **BIBLIOGRAPHY**